

Thy Tran

425-686-4214 | thytran.swe@gmail.com | linkedin.com/thytran1402 | github.com/ThyTran1402 | ttran142.com

EDUCATION

University of South Florida

May 2025

Bachelor of Arts in Computer Science GPA: 3.8

Coursework: Data Structures and Algorithms, Web Development, Databases, Object-Oriented Programming (OOP), Artificial Intelligence, Networking

EXPERIENCE

Google

May 2024 – Apr 2025

Google Software Engineering Fellow - Summer Program

- Selected among **80 fellows out of 6000 applicants** to participate in a **10-week mentorship**, focusing on advanced **algorithms, hands-on experience and system design** under mentorship from Google engineers
- Engineered a **task recommender** leveraging **Python, Docker, Vertex AI Embeddings**, and **BigQuery SQL** to rank personalized study tasks, improving model retrieval latency by **35%** during mentor-led performance reviews

Kogna AI

May 2025 - Present

Software Engineer (Cloud Infrastructure Team)

- Developed an **LLM-driven anomaly detector** using **Python, BigQuery ML**, and **Gemini API** to flag abnormal payment flows, reducing false positives by **32%** and caught **8000+** errors across enterprise transaction logs
- Developed scalable payment APIs using **.NET Core C#**, orchestrated via **Kafka Streams** and **SQL replication**, decreasing system latency by **20ms** and reducing error rate by **40%** across regional clusters
- Automated cloud observability pipeline with **Terraform, Prometheus**, and **Grafana on Google Kubernetes Engine (GKE)**, cutting manual incident response time by **60%** through unified alerting and metric dashboards

OpenSearch Project

Dec 2025 – Present

Open Source Contributor: [opensearch-k8s-operator](#) & [opensearch-build](#)

- Fixed data serialization and API migration bugs in a **Go-based Kubernetes** operator, preventing runtime failures across **1,000+** deployments
- Built automated CVE scanning into **CI/CD** pipelines, reducing security vulnerabilities in release builds
- Fixed runtime errors in remote artifact retrieval and added unit test coverage for multi-cluster integration test scenarios (cross-cluster replication, security/non-security configs)
- Redesigned cluster log resolution in **OpenSearch's Python-based CI/CD** framework, replacing hardcoded ID mappings with dynamic manifest-driven computation, supporting multi-node topologies across **2+ cluster** configs per component

University Of South Florida

Apr 2024 – Present

Teaching Assistant, COT 4400: Analysis of Algorithms

- Taught and led 10 weekly lab sections for **20 students**, covering **C++ data structures** including arrays, linked lists, stacks, queues, trees, and graphs through hands-on coding exercises and problem-solving sessions
- Collaborated with professors to design and grade **C++ algorithmic assignments**, provided inline feedback via **GitHub Classroom**, and maintained **unit tests and autograders** to ensure correctness and performance

Avocademy Inc (YC W22)

May 2024 – Aug 2024

Software Engineer Intern (Web Development Team)

- Built internal **analytics dashboard** using **React.js, Redux**, and **Bootstrap**, allowing **teaching teams** to visualize student progress and performance trends, improving reporting efficiency by **40%** and adopted by **100+ instructors**
- Enhanced backend for data history in **Golang, MySQL**, and **Azure Blob Storage**, implementing **timestamped record logging** and **secure file retrieval API**, increasing internal tool engagement by **80%**
- Implemented **40+ unit and integration tests** in **Go** using **Testify** and **Ginkgo**, raising test coverage to **85%** and reducing manual QA time by **20%** in weekly CI/CD deployments on Github Actions

PROJECTS

Carbonator (Top 1 ShellsHack) | *Flask, Arduino, Azure IoT Hub, CosmosDB, TensorFlow* Sep 2023 - May 2024

- Built a **real-time greenhouse monitoring platform** using **Azure IoT Hub, CosmosDB**, and **Flask** to collect and visualize **547+ sensor readings**, enabling early detection of soil humidity and temperature anomalies
- Trained and deployed a lightweight **CNN model** on **OpenMV microcontroller** with **85% accuracy** in crop-weed classification, improving automated plant health detection efficiency by **40%**

TECHNICAL SKILLS

Languages: C++, Python, Java,, C#, TypeScript, JavaScript, Go, C, Bash, HTML/CSS, SQL

Frameworks/Tools: .NET, React.js, Next.js, Spring Boot, Flask, FastAPI, Node.js, Express, Tailwind, Redux, gRPC

Databases: DynamoDB, PostgreSQL, MongoDB, Redis, MySQL, Supabase

Cloud/Infra: AWS (ECS, S3, RDS, Lambda), GCP, Docker, Kubernetes, GitHub Actions, Terraform, Linux

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, LangChain, OpenAI API